



13. How can a parent process wait for a child process to complete in UNIX?
 - A) pp()
 - B) wait()
 - C) pause()
 - D) child()
14. Which is the core of the operating system?
 - A) Shell
 - B) Kernel
 - C) Command
 - D) Script
15. Which command is used to insert a kernel module into the Linux kernel?
 - A) modprobe
 - B) insmod
 - C) loadmod
 - D) insertmod
16. What command is used to remove a kernel module?
 - A) rmmod
 - B) delmod
 - C) unloadmod
 - D) rmkernel
17. Which function is called when a Linux kernel module is loaded?
 - A) module_load()
 - B) kernel_init()
 - C) module_init()
 - D) init_module()
18. Which function is executed when a kernel module is removed?
 - A) module_exit()
 - B) cleanup_module()
 - C) exit_module()
 - D) unload_module()
19. Which command is used to display the operating system name
 - A) uname
 - B) os
 - C) kernel
 - D) boot
20. What is the purpose of the MODULE_LICENSE() macro in a kernel module?
 - A) Specifies the module's file permissions
 - B) Defines the module's compatibility with the kernel
 - C) Declares the module's license type (e.g., GPL)
 - D) Registers the module's author information

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) What are the functionalities of cat command in Unix? Given an example to demonstrate each functionality. | 8 | L*2 | 1 | 1 |
| b) Devise a shell script to accept two files as positional parameters and display the content of the file that has maximum size. Display the interactive messages to the user if file not found or missing parameter. | 8 | L3 | 2 | 1 |
| 2. a) How exist status of any Unix command is verified? Illustrate the usage of exist status with an example script | 8 | L2 | 1 | 1 |
| b) Devise a shell script which accept two integer values as input arguments and perform the following arithmetic operation based on the option selected by the user. | | | | |

Options	Arithmetic operation
1	Addition
2	Subtraction
3	Multiplication
4	Division

- | | | | | |
|---|---|----|---|---|
| 3. a) Write a Unix shell script to demonstrate the following programming features with an example | 8 | L3 | 2 | 1 |
| i) case statement ii) if statement | 8 | L2 | 1 | 1 |

- b) How unix will handle positional parameters? Illustrate the way of handling positional parameters with example script.

8 L3 2 1

Unit – II

4. a) How Linux kernel is organized in Linux OS distribution? Explain the functionality of each block.

8 L2 3 1

- b) Write a C code to search a given string in a file. Use system call and display the frequency of the string. Declare a user defined function to find the frequency
int search_file_string(int file_handler, char *string_value, int length).

8 L3 4 1

5. a) With an example, illustrate the following Unix system calls
i) open ii) write

8 L2 3 1

- b) What is U-Boot? Explain its role in embedded Linux systems and how it differs from other bootloaders

8 L2 4 1

6. a) Write a C code using system call to read two numbers. Perform the addition and save the result in to a file. The file should contain the values of the two numbers along with sum. Pass the file name as positional parameter. Prompt the necessary error message to user.

8 L3 3 1

- b) What is the role of bootloader in a Linux system? Explain its importance in the booting process.

8 L2 4 1

Unit – III

7. a) Make a LKM module to access a IO port considering raspberry Pi example

8 L3 5 1

- b) Illustrate the use of following commands with an example

i. lsmod

ii. insmod

8 L2 5 1

8. a) Compare and contrast character device drivers and block device drivers with examples.

8 L2 5 1

- b) List the procedure to compile a Linux kernel module with an example.

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
